LESSON 1.3

Math is In My World

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can represent a real-world situation with a tape diagram or table and use decimal operations to solve it.

WHY THIS MATTERS Math lives in your world — in coins on a table and tram cars over a valley. Today you visualize real situations, pick tools like tape diagrams and tables, and let decimal arithmetic do the heavy lifting. The tram must carry 4,000 passengers over the valley. Each ride carries 80 passengers and takes 12.5 minutes. How many rides are needed, and how long will all of the rides take?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
representation *Spanish: representación*	A picture, diagram, table, or equation that shows the mathematics in a situation.	The same 12 shown three ways: a tape diagram, a table, and 3 × 4.
tape diagram *Spanish: diagrama de cintas*	A rectangle cut into equal parts that helps you visualize how a total breaks apart.	One bar of 4,000 passengers cut into equal groups of 80.
tool *Spanish: herramienta*	Anything you choose to help you see relationships and solve a problem — a table, a diagram, a graph.	A table, a number line, or a tape diagram — whichever shows the relationship.
ordered pair *Spanish: par ordenado*	Two numbers that go together, like a time and a number of trips, that you can plot as a point.	(4, 60): 4 hours costs $60 — the hours always come first.
round trip *Spanish: viaje de ida y vuelta*	A complete journey from the start, to the destination, and back to the start.	5 miles out plus 5 miles back = 10 miles in all.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A pile of coins is too big to count one at a time. About how many coins are in the pile?

1. I do not count every coin — I find a small stack I CAN count, about 10 coins.
2. Then I estimate how many stacks like that are in the pile. I count about 15.
3. 15 stacks of about 10 is 15 × 10 = 150, so my estimate is about 150 coins — reasoned, not random.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

One tram ride takes 12.5 minutes, so 10 rides take 12.5 × 10 = 125 minutes and 50 rides take 12.5 × 50 = 625 minutes.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- This situation needs _____ because we are _____ a total, not _____ it.
- First I counted _____ .
- Then I estimated about _____ because _____ .
- My tape diagram shows _____ .

Word bank representation | tape diagram | tool | ordered pair | round trip

WATCH OUT!

A common mistake

Converting minutes to hours by moving the decimal point

LESSON 1.3

Math is In My World

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

One tram ride takes 12.5 minutes. How many minutes do 10 rides take?

(A) 1.25 minutes
(B) 22.5 minutes
(C) 125 minutes
(D) 1,250 minutes

HINT You need 10 equal groups of 12.5.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Five Breaks: which of these is a correct way to decompose 12.5?

(A) 1 + 2.5
(B) 10 + 2.5
(C) 10 + 2.05
(D) 12 + 5

HINT Add each option and compare to 12.5.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

How many minutes do 50 tram rides take, at 12.5 minutes per ride?

(A) 62.5 minutes
(B) 600 minutes
(C) 625 minutes
(D) 6,250 minutes

HINT Use the table pattern: 10 rides take 125 minutes.

PER 1	

1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

4 **Circle the letter of the best answer. Show how you know.**

Each tram ride carries 80 passengers. How many rides are needed to transport 4,000 passengers?

(A) 50 rides
(B) 500 rides
(C) 3,920 rides
(D) 320,000 rides

HINT You are splitting 4,000 into equal groups of 80.

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

5 **Answer in complete sentences.**

A coin display has about 15 stacks that look the same size. You count 10 coins in one stack. About how many coins are in the display? Explain how to estimate without counting every coin.

SENTENCE STARTER First I counted ___ .
Then I estimated about ___ because ___ .

WRITE YOUR ANSWER

6 **Answer in complete sentences.**

A tram carries 80 passengers per ride. How can a tape diagram show how many rides transport 4,000 passengers?

SENTENCE STARTER My tape diagram shows ___ .
It helps because ___ .

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Find a card where you and your partner chose different operations. What is it about the situation — not the numbers — that tells you whether division or multiplication helps?

SENTENCE STARTER This situation needs ___ because we are ___ a total, not ___ it.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 1.3

Math is In My World

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

The 50 rides take 625 minutes. About how many HOURS is that?

(A) About 6.25 hours
(B) About 10.42 hours
(C) About 12.5 hours
(D) About 62.5 hours

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

How does the answer change for 2,000 passengers? Each ride carries 80 passengers and takes 12.5 minutes. How many MINUTES are needed?

(A) 31.25 minutes
(B) 300 minutes
(C) 312.5 minutes
(D) 3,125 minutes

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

One round trip takes 35 minutes. Using the table (35 → 1, 350 → 10, 525 → 15, 700 → 20), how many round trips does a tram car complete in 10 hours?

(A) 15 round trips
(B) 17 round trips
(C) 18 round trips
(D) 20 round trips

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

The round trip is 5 miles. How far does a tram car travel in 5 round trips?

(A) 1 mile
(B) 10 miles
(C) 25 miles
(D) 50 miles

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

5 **Answer in complete sentences.**

Compare these tram problems. A: A tram carries 80 passengers per ride and each ride takes 35 minutes. How long would 4,000 passengers take? B: A different tram completes a round trip in 12.5 minutes. How many round trips fit in 10 hours? How are the problems alike and different?

WRITE YOUR ANSWER

6 **Answer in complete sentences.**

How does the answer change if the tram must transport 6,000 passengers? (Each ride carries 80 passengers and takes 12.5 minutes.) Show your steps.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Find a card where you and your partner chose different operations. What is it about the situation — not the numbers — that tells you whether division or multiplication helps?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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LESSON 1.3

Math is In My World

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each tram question to the operation that answers it.

_____	Minutes for 40 rides (12.5 min each)	A	40 × 12.5
_____	Rides needed for 400 passengers (80 per ride)	B	400 ÷ 80
_____	Hours in 625 minutes	C	625 ÷ 60
_____	Passengers on 30 full rides	D	30 × 80

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Apply — What Time Is It? The new year starts at midnight on January 1st. What time will it be 2,022 minutes after midnight?

(A) 8:22 p.m. on January 1
(B) 9:42 a.m. on January 2
(C) 9:42 a.m. on January 1
(D) 9:42 p.m. on January 2

SHOW YOUR WORK

3 Circle the letter of the best answer. Show how you know.

How many rides would the tram need for 10,000 passengers, at 80 passengers per ride?

(A) 12.5 rides
(B) 125 rides
(C) 250 rides
(D) 1,250 rides

PER 1

PART 3 Explain and correct

reasoning

4 Answer in complete sentences.

The lesson says we make strategic decisions about which tool to use. For the 10-hour round-trip question, would you choose a table of values, a coordinate plane, or an equation? Defend your choice.

WRITE YOUR ANSWER

5 Answer in complete sentences.

Math is in YOUR world. Describe a real situation from your life that works like the tram problem — a total split into equal groups, or a time built from equal chunks — and show the math.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Find a card where you and your partner chose different operations. What is it about the situation — not the numbers — that tells you whether division or multiplication helps?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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